Applying Data Binding and MVVM

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Module Outline

Learn the data binding basics

- Use the Binding markup extension
- Bind to another element
- Know how the data context works

Apply the MVVM pattern

Notify about property changes

Use the x:Bind markup extension

- Differences between Binding and x:Bind

Demo

Bind to another element

Know How the Data Context Works

ElementName

```
<TextBlock Text="{Binding ElementName=...}"/>
```

Know How the Data Context Works

ElementName

Source

```
<TextBlock
  Text="{Binding Source={StaticResource myRes}}"/>
```

Know How the Data Context Works

ElementName **Source** **RelativeSource**

```
<Rectangle Fill="Red" Width="300"
  Height="{Binding RelativeSource=
            {RelativeSource Self},Path=Width}"/>
```

Know How the Data Context Works

ElementName

Source

RelativeSource

DataContext

Know How the Data Context Works

```
<TextBlock Text="{Binding}"/>
```

DataContext

Know How the Data Context Works

```
<Grid>
  <StackPanel>
    <TextBlock Text="{Binding}"/>
  </StackPanel>
</Grid>
```

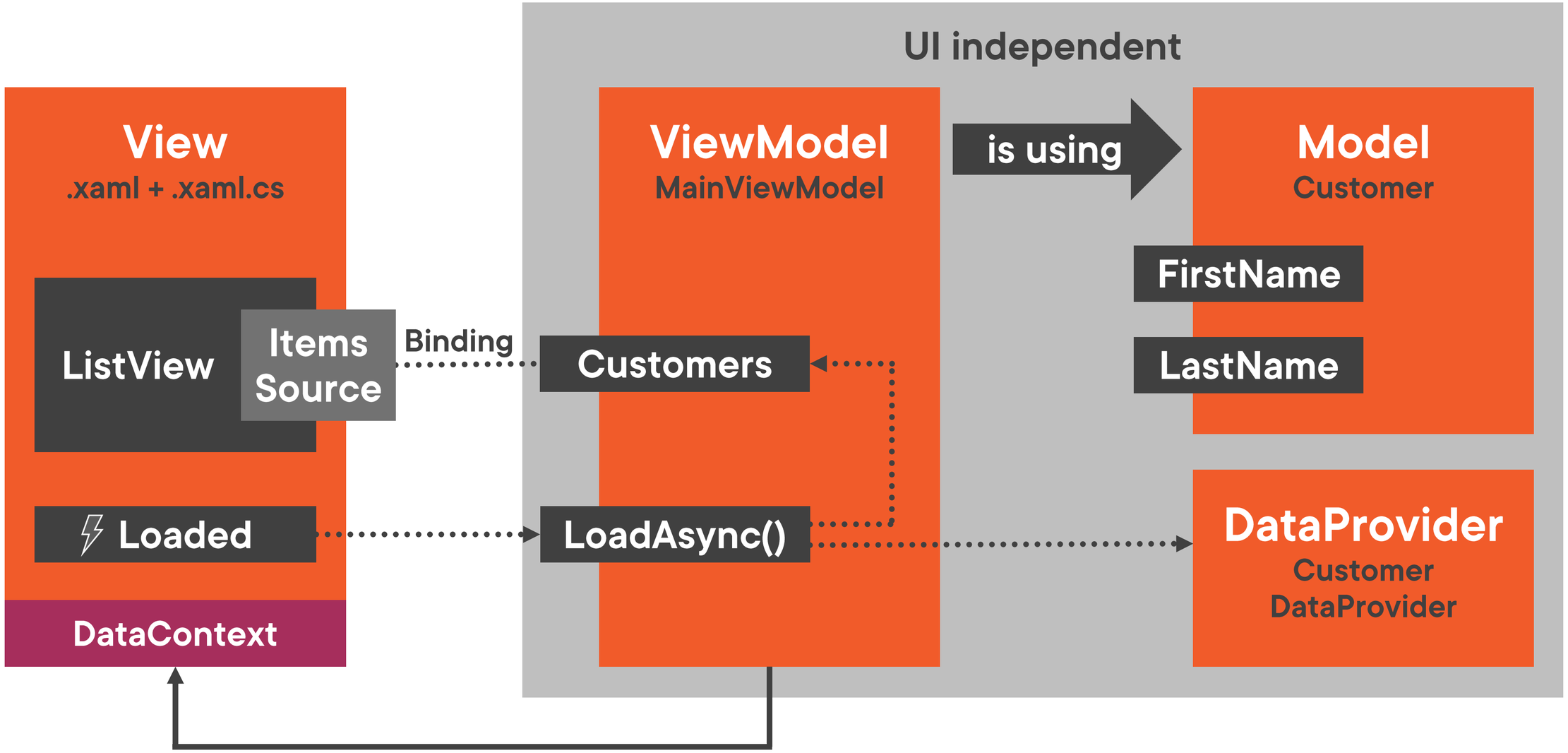
Know How the Data Context Works

```
<Grid DataContext="Pluralsight">
  <StackPanel>
    <TextBlock Text="{Binding}"/>
  </StackPanel>
</Grid>
```

Know How the Data Context Works

```
<Grid DataContext="Pluralsight">
  <StackPanel DataContext="Thomas">
    <TextBlock Text="{Binding}"/>
  </StackPanel>
</Grid>
```

The Model View ViewModel (MVVM) Pattern



Advantages of MVVM

Maintainability

Testability

Advantages of MVVM

Maintainability

Testability

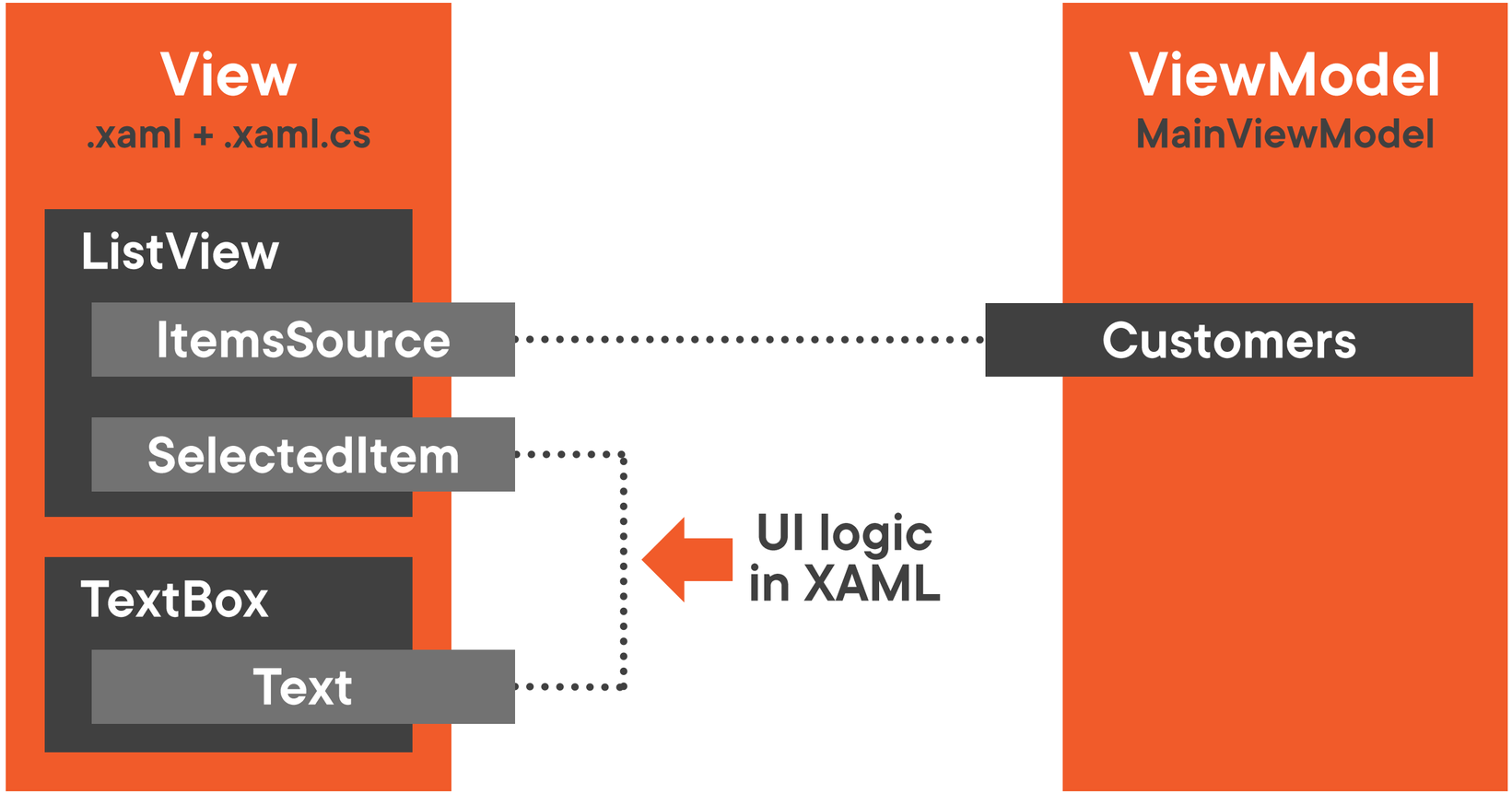
Demo

Create a MainViewModel

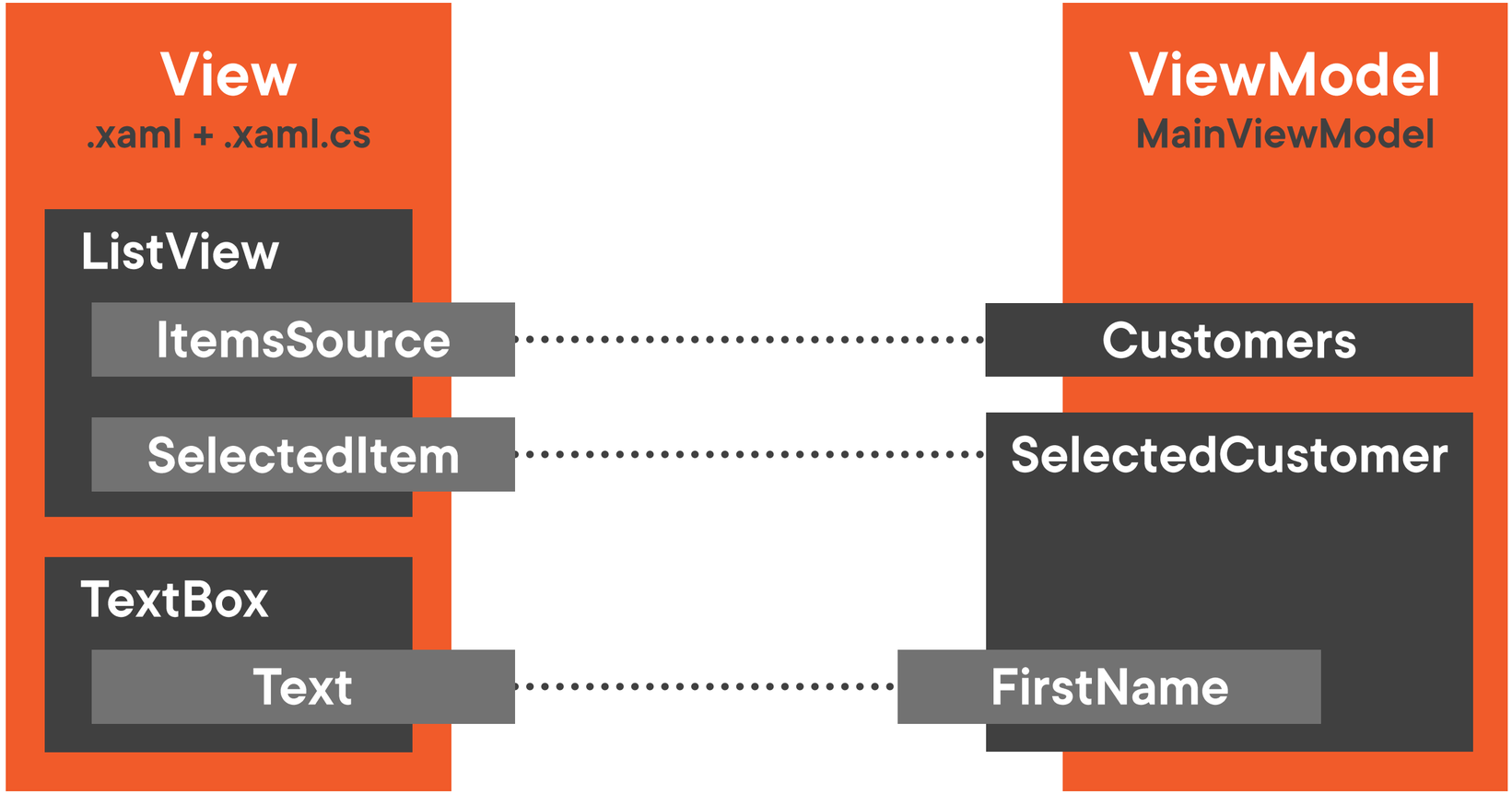
Demo

Use the MainViewModel for the MainWindow

Add a SelectedCustomer Property



Add a SelectedCustomer Property



Demo

Add a SelectedCustomer property

Demo

Notify about property changes

Demo

Refactor logic into a ViewModelBase class

Demo

Create a CustomerItemViewModel

Use x:Bind Instead of Binding

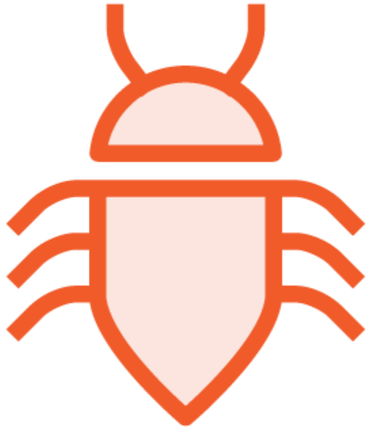
```
<TextBlock Text="{Binding Path=...}"/>
```

```
<TextBlock Text="{x:Bind Path=...}"/>
```

Advantages of x:Bind

Better performance

Compile-time errors



Better debugging experience

Prefer x:Bind over Binding whenever possible

Understand the Data Source of x:Bind

{Binding}

- ElementName
- Source
- RelativeSource
- DataContext

{x:Bind}

- Root object of the XAML document

Instance of the class specified with the x:Class attribute

Demo

Use x:Bind instead of Binding

Demo

Bind the Visibility with x:Bind

Summary

Create data bindings in XAML

- Use the Binding markup extension
- Bind to another element
- Know how the data context works

Apply the MVVM pattern

- Notify about property changes

Use the x:Bind markup extension

- Prefer x:Bind over Binding
- Compile-time errors

Up Next:
Executing Code with Commands